Nallabati Venkata Durga Sai Sarat Chandra

Target Role: Data Analyst Intern | **Swiggy**

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SUMMARY

Visionary Robotics & AI undergraduate specializing in the intersection of edge mechatronics, Scientific Machine Learning, and human-centric design. Combines hands-on development of autonomous hardware and advanced physics-informed models with intuitive UI/UX architecture. Poised to deliver high-impact, user-friendly edge intelligence systems that seamlessly bridge complex physical dynamics with operational clarity.

EDUCATION

Bachelor of Technology (B.Tech), Amrita Vishwa Vidyapeetham — **2024 – Present**
CGPA: 9.05 — *Amritapuri*

Higher Secondary Education, Sri Chaitanya Vidhyanikethan — **2024**
CBSE (MPCEA) Percentage: 88.2% — *Vishakapatnam*

Secondary School Education, Sri Chaitanya School — **2022**
SSC AP Marks: 570/600 — *Rajam*

SKILLS

Core Stack & Systems: Python, C, C++, SQL, Git, Linux, Ubuntu Server, ROS 2
AI, ML & SciML: Physics-Informed Machine Learning (PIML), Deep Learning, Optuna (Bayesian Optimization), Explainable AI (SHAP)
Data Analytics & Modeling: Pandas, NumPy, Scikit-Learn, LightGBM, XGBoost, Conformal Prediction, Time-Series DSP
Engineering & Simulation: Computational Fluid Dynamics (ANSYS Fluent), Thermodynamics, CAD (Fusion 360, AutoCAD)
Design & UI/UX Tools: Figma, Adobe Photoshop, Canva, UI/UX Wireframing, Responsive Layouts

EXPERIENCE

InAmigos Foundation (Internship), *Graphic Designer* — **06/2025**

- Developing visual content, including posters and reels, for various organizational programs to support fundraising initiatives and enhance public engagement across social media platforms. Utilized Adobe Photoshop, Canva and some AI tools to produce engaging visual assets.

PROJECTS

Evolution Edge: Self-Evolving Neural Bridge — **05/2026**

- Engineered a hybrid AI pipeline utilizing ROCm 6.x and ONNX Runtime to orchestrate low-latency inference on Ryzen AI NPUs with symbolic confidence-based escalation to AMD Instinct MI300X clusters. Implemented persistent on-device learning through knowledge distillation and PEFT-driven weight updates, bridging the gap between edge efficiency and cloud-scale intelligence.

Physics-Informed Multi-Modal Industrial Diagnostic Framework — **05/2026**

- Engineered a Physics-Informed Machine Learning (PIML) framework combining mechanical heuristics and data-driven models to predict equipment failures. Optimized with SMOTE and Optuna for an 85.2% recall and integrated with SHAP for actionable transparency, the simulated system demonstrated a 68% reduction in maintenance costs, saving an estimated $230,000 across a 2,000-machine fleet.

AI-Driven Hemodynamic Digital Twin & Rheology Discovery — **01/2026**

- Developed a Physics-Informed Neural Network (PINN) framework to autonomously discover non-Newtonian blood rheology parameters from pulsatile CFD datasets of stenosed arteries. Engineered a high-accuracy hybrid pipeline integrating multi-fidelity data with boundary condition reconstruction, achieving <2% error in predicting complex hemodynamic flow fields.

GSM-Based Smart Switch for Industrial Automation, *Prof. Sumesh K.P* — **12/2025**

- Developed a compact remote-automation node utilizing the GSM protocol for managing industrial loads in off-grid areas without internet access. Created a reliable asynchronous parser for SMS-based interrupts, ensuring over 99% uptime for remote hardware resets and power.

Multi-Parameter Health Vitals Monitoring and Real Time IoT Data Streamer, *Prof. Amal.P* — **11/2025**

- Designed a clinical monitoring node with ESP8266 for biometric data acquisition and cloud telemetry. Integrated MAX30100 and DHT11 sensors, using digital signal processing for heart rate and SpO2 extraction. Created a low-latency interface via Blynk IoT for remote patient monitoring.

PUBLICATIONS

Small Scale Raspberry Pi Based Autonomous Car: A Practical Guide to Understand Complexity, *IEEE ICRM 2025* — **11/2025**

- Architecting Level 4 autonomous navigation logic within a 2.4 GHz CPU hardware bottleneck, utilizing Probabilistic Hough Transforms for real-time computer vision processing, and engineering a handshake-free UDP telemetry protocol to eliminate multi-agent V2V latency in edge environments.

CLUB EXPERIENCE

Embedded Systems and Robotics Developer, *CLUB MIRAI* — **12/2025 – Present**

- Utilising ROS 2 Jazzy to create a low-latency distributed control system for actuator and perception nodes. Developing a 2D/3D kinematic model for a companion robot and applying inverse kinematics (IK) for motor joint angles. Implementing optimised libraries for high frame-rate detection in a dynamic face tracking computer vision pipeline. Establishing a Raspberry Pi 5 remote development environment on Ubuntu Server for HIL testing and headless ROS 2 execution.

ACHIEVEMENTS

- Selected among top international teams to complete all three technical milestones of the IEEE IES Generative AI Challenge.
- Successfully cleared the 2 Levels of NAEST (India's premier physics experimental competition organized by IAPT and Dr. H.C. Verma).
- Selected for the ISRO-IIRS YUVIKA through a national-level merit screening and science aptitude evaluation.
- Qualified for MTSO (Maths Olympiad) Level 2 (Stage II) by ranking in the top tier of candidates nationwide.